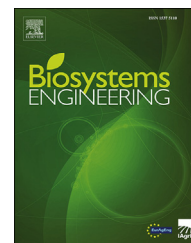


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Research Paper

Dynamic and explainable fish mortality prediction under low-concentration ammonia nitrogen stress



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Fish mortality prediction is important in intensive aquaculture; however, it is difficult to accurately predict fish mortality using statistical prediction methods because of dynamic changes in environmental factors. Therefore, in this study, dynamic fish mortality prediction under environmental stress due to low concentrations of ammonia nitrogen was performed on the basis of 32 features, including static and cumulative time-series features. First, the effects of different ammonia nitrogen concentrations on fish mortality were investigated, and the median lethal concentrations of ammonia nitrogen at 24, 48, 72, and 96 h were determined. Thereafter, a machine learning model was proposed to predict the hourly mortality of fish under stress due to low-concentration ammonia nitrogen. The proposed mortality prediction model combines static data with cumulative time-series dynamic data and uses a long short-term memory artificial neural network for dynamic fish mortality prediction. To confirm the utility of the model and improve its interpretability, the features driving fish mortality prediction were visualised at 1-h intervals. The model showed excellent predictive performance on the external validation dataset, which improved over time, achieving a coefficient of determination (R^2) of 0.81 (95% confidence interval (CI) 0.73–0.89) and root-mean-square error (RMSE) of 0.30 (CI 0.21–0.39) within 168 h. This dynamic predictive method adapts to changing aquaculture conditions and provides interpretable guidance for aquaculturists.

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1. Introduction

Fish mortality prediction is important in intensive aquaculture environments (Andersen, 2020; Fisher et al., 2021; Lopes et al., 2021). Several prediction methods such as score-based prediction using life history parameters (Thorson et al., 2017), mark-recapture methods (Kramer et al., 2019; Lees

et al., 2021), age structure analysis (Vaz-Dos-santos & da Silveira, 2021), and prediction models based on fish lethality functions have been used to predict fish mortality (Ewees et al., 2021; Walters & Christensen, 2019). However, their low applicability has hindered their widespread use. Most existing mortality prediction methods are static (Lopes et al., 2021) because they are based on data obtained from the physical parameters of fish at a given moment in time. Owing to the

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Nomenclature			
Symbols			
K	Ionisation constant, mol L ⁻¹	SC	Safe ammonia nitrogen concentration, mg L ⁻¹
T_k	Kelvin temperature of water, K	T	Water temperature, °C
h_{t-1}	Hidden state of the previous layer	DO	Dissolved oxygen, mg L ⁻¹
x_t	Current input data	pH	Hydrogen ion concentration, mol L ⁻¹
f_t	Forgetting gate	NH ₃	Inorganic nitrogen, mg L ⁻¹
w_f	Weights of forgetting gate	TAN	Total ammonia nitrogen, mg L ⁻¹
b_f	Biases of forgetting gate	96-h TAN LC ₅₀	Median lethal concentration of total ammonia nitrogen within 96 h, mg L ⁻¹
$\tilde{C}_t$	Candidate vector of the current cell state	AVE_w	Average weight of the fish, g
i_t	Input gate	AVE_l	Average length of the fish, cm
w_i	Weights of input gate	AVE_a	Average monthly age of the fish, month
b_i	Biases of input gate	CD	Culture density, kg m ⁻³
w_c	Weights of candidate vectors	EC	Electrical conductivity, S m ⁻¹
b_c	Biases of candidate vectors	TUR	Turbidity of culture water, FTU
C_{t-1}	Cell state of the previous layer	TN	Total nitrogen, mg L ⁻¹
C_t	Updated cell state	TP	Total phosphorus, mg L ⁻¹
h_t	Hidden state of the current layer	RC	Residual chlorine of culture water, mg L ⁻¹
O_t	Output gate	R ²	Coefficient of determination
w_o	Weights of output gate	TAN LC ₅₀	Median lethal concentration of total ammonia nitrogen, mg L ⁻¹
b_o	Biases of output gate	CI	95% confidence interval
ϕ_i	Shapley value for each feature	DO_max	Dissolved oxygen maximum, mg L ⁻¹
f	Prediction model	DO_med	Dissolved oxygen median, mg L ⁻¹
x	Input sample data	DO_min	Dissolved oxygen minimum, mg L ⁻¹
z'	Subset of feature combinations	pH_max	Hydrogen ion concentration maximum, mg L ⁻¹
x'	Simplification of the input data	T_med	Water temperature median, °C
M	Total number of features	T_max	Water temperature maximum, °C
N	Number of samples	pH_diff	Hydrogen ion concentration difference, mg L ⁻¹
y_i	True value	Abbreviations	
$\hat{y}_i$	Predicted value	LSTM	Long short-term memory
$\bar{y}$	Sample mean	PCA	Principal component analysis
LC ₅₀	Median lethal concentration, mg L ⁻¹	RF	Random forest
24-h TAN LC ₅₀	Median lethal concentration of total ammonia nitrogen within 24 h, mg L ⁻¹	RMSE	Root-mean-square error
NH ₃ LC ₅₀	Median lethal concentration of inorganic nitrogen, mg L ⁻¹	SHAP	Shapley additive explanations

lack of adequate measures to make time-series predictions, breeders may resort to subjective judgements, and even experienced culturists are prone to bias. In addition, fish in actual culture ponds exist under low-concentration ammonia nitrogen stress for long periods (Elbially et al., 2021; Kathayyani et al., 2019), and environmental factors that significantly affect fish can easily be overlooked in the prediction.

Modern methods of recirculating water culture allow for the continuous measurement of several environmental parameters, resulting in a significant amount of culture data, which include valuable environmental information about fish survival outcomes (Nys et al., 2020; Saeed et al., 2022; Suwarsito et al., 2020). These data are often heterogeneous, consisting of irregular sampling, artefacts, and varying degrees of completeness of structured and unstructured information; this pose a challenge to traditional statistical models. Machine learning provides an effective solution to this problem, as it is capable of obtaining vital information from complex data and contributing to the decision-making

process (Wu et al., 2022). A recent study used intensive care unit data to build a machine-learning-based mortality prediction model for patients, which outperformed traditional methods (Alves et al., 2019; Johnson et al., 2016; Nielsen et al., 2019). The advantages of complex models outweigh those of simple models; however, they also pose some disadvantages. The internal mechanisms of these models are complex and estimating the importance of each feature is difficult even when viewing all model parameters. Recently, methods have been proposed to explain specific predictive models, which can help stakeholders understand why the model makes certain decisions or predictions (Fintz et al., 2022; Ribeiro et al., 2016). This facilitates the use of complex models for production decisions.

In this study, we first performed an ammonia nitrogen toxicity test to investigate the effects of different ammonia nitrogen concentrations on fish, and determined the median lethal concentrations (LC₅₀) of ammonia nitrogen for fish at 24, 48, 72, and 96 h. Thereafter, we used the proposed machine

learning model to predict the hourly mortality of fish under low-concentration ammonia nitrogen stress. The proposed mortality prediction model combines static data with cumulative time-series dynamic data and uses a long short-term memory (LSTM) artificial neural network to predict dynamic fish mortality. Furthermore, to confirm the utility of the model and improve its interpretability, we visualised the features driving fish mortality prediction in individual culture pools at 1-h intervals.

Notably, this study devised a dynamic and interpretable method for predicting mortality in fish under low-concentration ammonia nitrogen stress. The key contributions of this study are as follows:

- (1) In aquaculture, fish are prone to low-ammonia stress. Hence, an ammonia nitrogen toxicity test was conducted to determine the LC_{50} and safe concentration (SC) of ammonia nitrogen in American black bass (*Micropterus salmoides*).
- (2) For complex and dynamically changing aquaculture environments, LSTM was used to predict the mortality of American black bass under low-concentration ammonia nitrogen stress.
- (3) To alleviate the black box problem of deep learning models, we propose the use of game theory to visualise the effects of the features on the prediction result and improve the interpretability of the prediction model.

The remainder of this paper is organised as follows. Section 2 presents the ammonia nitrogen toxicity test and predictive model design. In Section 3, the performance of the proposed method is described to demonstrate the accuracy and interpretability of the proposed prediction model. Section 4 discusses the results and the advantages of the proposed method. Finally, Section 5 summarises our findings and outlines the limitations of the study.

2. Materials and methods

Ethical approval

All procedures in this study were conducted according to the 'Guiding Principles in the Care and Use of Animals' (China) and approved by the Experimental Animal Ethics Committee of Nanjing Agricultural University (NJAU.No20200408072).

2.1. Ammonia nitrogen toxicity test

2.1.1. Fish

American black bass (*M. salmoides*), a fish belonging to the genus Sunfish, was purchased from a fish fry breeding base in Guangzhou, China, and temporarily reared in a recirculating water culture system at a water temperature (T) of $21.0\text{ }^{\circ}\text{C}$, dissolved oxygen (DO) of 8.0 mg L^{-1} , and hydrogen ion concentration (pH) of 7.9. During the temporary culture period, the fish were fed twice a day at 8:00 a.m. and 8:00 p.m., and one-third of the water volume was changed daily, with continuous oxygenation and regular cleaning of faeces and

residual bait for seven days. Healthy individuals (Endo & Wu, 2019) with a weight of $4.0 \pm 0.2\text{ g}$ were selected as the experimental subjects, and feeding was suspended 24 h before the experiment.

2.1.2. Ammonia nitrogen toxicity test

The experimental water used to achieve the required ammonia nitrogen concentration was prepared using 10 g L^{-1} of analytically pure ammonium chloride (NH_4Cl , 99.5%) mother liquor. An ammonia nitrogen toxicity test was conducted to determine the LC_{50} of juvenile American black bass under stress conditions at 24, 48, 72, and 96 h. The actual ammonia nitrogen concentration in the water was measured every 6 h using Nessler's reagent spectrophotometry (Wu & Cao, 2013), and the concentration was adjusted periodically to maintain the ammonia nitrogen concentration in the fish tank during the initial measurement. The inorganic nitrogen (NH_3) concentration was determined using the total ammonia nitrogen (TAN) concentration as follows:

$$\text{NH}_3 = \frac{\text{TAN}}{(10^{K-\text{pH}} + 1)}, \quad (1)$$

$$K = 0.09018 + \frac{2729.92}{T_k}, \quad (2)$$

where K denotes the ionisation constant, and T_k is the Kelvin temperature of water at the time of measurement. The SC for fish is expressed as follows:

$$\text{SC} = 10\% \times 96 - \text{h TAN } LC_{50}, \quad (3)$$

where $96 - \text{h TAN } LC_{50}$ represents the median lethal concentration of TAN within 96 h.

The experiment was set up with an interval concentration gradient of 0.25 mg L^{-1} of TAN. Three parallel groups, each with 20 fish, were established for each concentration. During the experiment, the mortality of the fish in the tank was observed and recorded every 1 h. The maximum stress concentration at which the fish did not die and the minimum stress concentration at which all the fish died within 24 h were counted. Six ammonia nitrogen stress groups and one experimental control group (0 mg L^{-1}) were designed for the above concentration range according to an equal logarithmic gradient. Three parallel groups were established for each group; ten fish were placed in each parallel group and subjected to 96 h of stress. The mortality rates of fish in different treatment groups were recorded at 24, 48, 72, and 96 h. The LC_{50} was determined using a probit regression approach (Gilenko & Chernova, 2021) at 24, 48, 72, and 96 h, based on the mortality rate.

2.2. Data acquisition for model building and validation

One hundred groups of culture ponds were established, and data were collected for American black bass under low-concentration ($\geq \text{SC}$ and $\leq 50\% \times (96\text{-h TAN } LC_{50})$) ammonia nitrogen stress. Weight, length, monthly age, and culture density of the fish in each group in the culture ponds were recorded. The water quality parameters were measured using sensors and spectrophotometry. Furthermore, feeding and disinfection operations of the culture ponds and fish mortality

were recorded every 1 h. The collected data were tested for correlation, and the data depicting high correlation with fish mortality ($P < 0.05$) were selected to constitute the static, dynamic, and operational data.

In this study, static data included the average weight of the fish (AVE_w), average length (AVE_l), average monthly age (AVE_a), and culture density (CD). The dynamic data were water quality monitoring data with high sampling frequency (sampling frequency of 1 min^{-1}) and water quality detection data measured using a spectrophotometer (sampling frequency of 6 h). Water quality monitoring data included T, DO, pH, and electrical conductivity (EC) of the culture pond. Water quality detection data include turbidity (TUR), total nitrogen (TN), total phosphorus (TP), residual chlorine (RC), TAN, and NH_3 in the culture water. Operational data were collected to record whether the feeding and sterilisation operations were performed in the culture pond every 1 h.

To optimise the obtained data, monitoring data with a high sampling frequency were aggregated into 1-h values (down-sampling), and features were constructed for the time series (upsampling) to produce five new features: minimum, maximum, median, average, and difference. The data collected every 6 h were unsampled into 1-h values using piecewise linear interpolation. Dummy coding was performed for non-binary categorical data (feeding, disinfection operations). Normalising all feature data to 0–1 ensured the stability of the features with negligible variance. Finally, the static, dynamic, and operational data were integrated to construct the original dataset for building the model. To validate the model, an additional 50 sets of breeding pools were established, and the operation was repeated to obtain an external validation dataset stored in the same format (Fig. 1).

2.3. Model development

In this study, a deep learning approach was applied to predict fish mortality within 168 h of low-concentration ammonia nitrogen stress (Fig. 2a). Specifically, LSTM, a temporal recurrent neural network was used (van Houdt et al., 2020). The LSTM neural network structure is suitable

for processing and predicting important events with very long intervals and delays in the time series. For example, the effects of various environmental variables on fish mortality in aquaculture are asynchronous and often exhibit time–accumulative properties. It is difficult to predict later events using traditional recurrent neural networks owing to the long time span of pre-order features. LSTM can solve the abovementioned problem by constantly cycling the information throughout the operation to ensure the persistence of the information.

The LSTM neural network structure requires time data series as the input to the model. We fused multiple sources of static, dynamic, and operational data from the breeding environment as inputs for the model. The patterns useful for the prediction task were then extracted from these time series data and used as input for the machine learning (Fig. 2b). The core of the LSTM neural network structure lies in the cell states and the ‘gate’ structure. The cell states are equivalent to the information transmission paths that allow information to be passed along in a sequence. It can be considered the ‘memory’ of the network. Theoretically, the cell states can pass on information relevant to the sequence processing throughout the duration of the process. Thus, even information from earlier time steps can be carried to cells at later time steps, overcoming the effects of short-term memory. Information is added and removed through ‘gate’ structures that learn whether to retain or forget an information during the training (Fig. 3).

Forgetting gate: This gate determines whether to retain or discard an information. The information from the previous hidden state h_{t-1} is passed to the sigmoid function along with the current input x_t , and the output value is between 0 and 1; a value closer to 0 means the information should be discarded, whereas a value closer to 1 means the information should be retained.

$$f_t = \sigma(w_f \cdot [h_{t-1}, x_t] + b_f), \quad (4)$$

where f_t denotes the forgetting gate, w_f and b_f are the weights and biases of the forgetting gate, h_{t-1} is the hidden state of the previous layer, and x_t denotes the current input data.

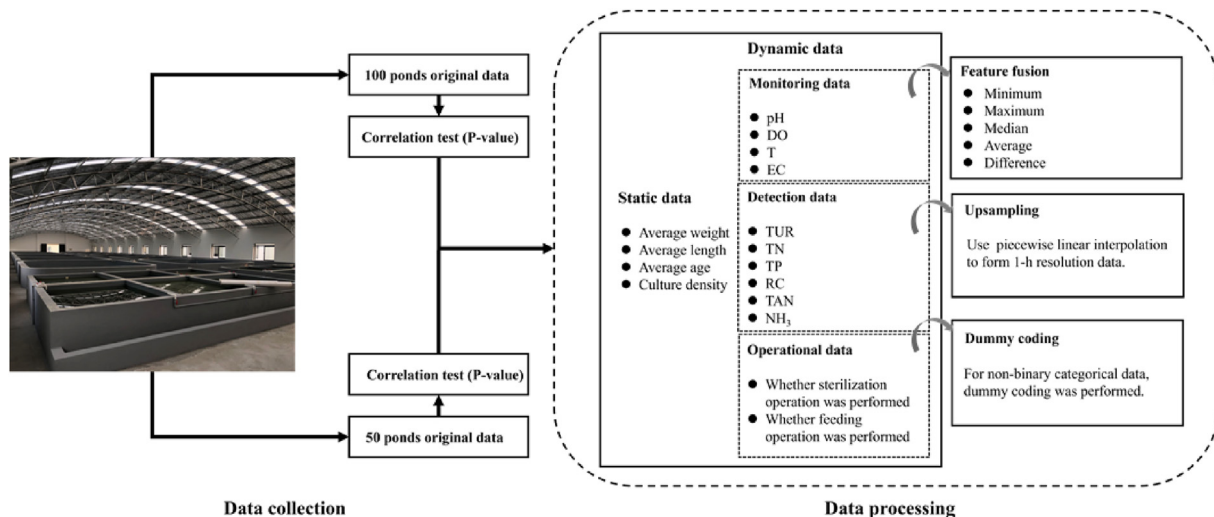
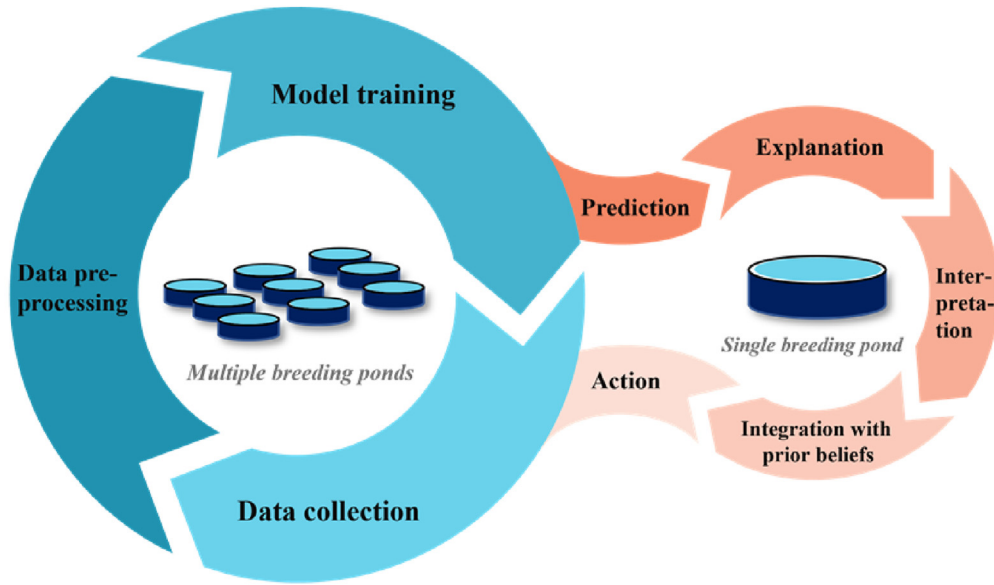
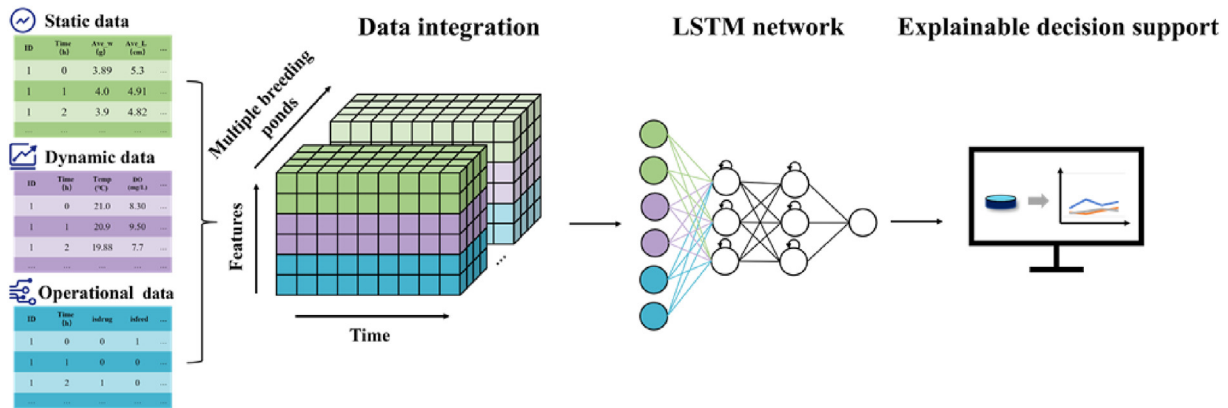


Fig. 1 – Data collection and processing.



(a)



(b)

Fig. 2 – Process overview of the proposed mortality prediction model.

Input gate: The input gate is used to update the cell state. First, the information from the previous layer of the hidden states h_{t-1} and the current input x_t are passed to the sigmoid function. The value obtained after the sigmoid function is adjusted between 0 and 1 to determine the information to be updated. A value of 0 indicates that the information is not important and 1 indicates that it is important. Next, the information from the previous layer of the hidden states h_{t-1} and the current input x_t are passed to the tanh function to create a new vector of candidate values $\tilde{C}_t$. Finally, the output of the sigmoid is multiplied by the output of tanh. The output of the sigmoid determines which information in the output of tanh is important and needs to be preserved.

$$i_t = \sigma(w_i \cdot [h_{t-1}, x_t] + b_i), \quad (5)$$

$$\tilde{C}_t = \tanh(w_C \cdot [h_{t-1}, x_t] + b_C), \quad (6)$$

where i_t denotes the input gate, w_i and b_i are the weights and biases of the input gate, respectively, $\tilde{C}_t$ is the candidate vector of the current cell state, and w_C and b_C are the weights and biases of the candidate vectors, respectively.

Cell state: The cell state of the previous layer C_{t-1} is multiplied point-by-point with the forgetting vector f_t . Multiplication with a value close to 0 indicates that the information in the new cell state is to be discarded. This value is then added point-by-point with the output value of the input gate to update the new information found by the neural network into the cell state. At this point, the updated cell state is obtained.

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad (7)$$

where C_t is the updated cell state and C_{t-1} is the cell state of the previous layer and $\odot$ represents the Hadamard product.

Output gate: The output gate is used to determine the value of the next hidden state that contains the information from

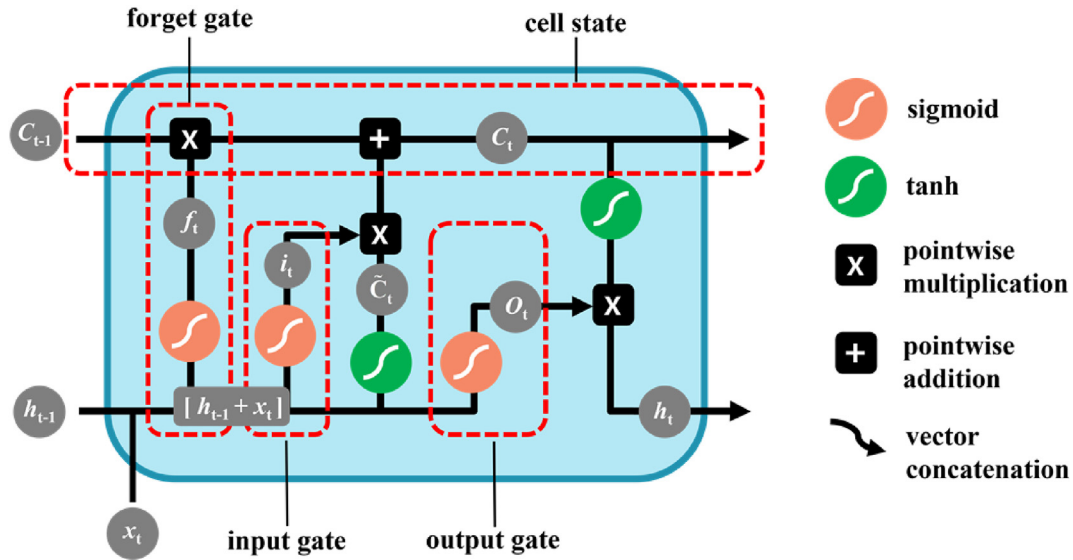


Fig. 3 – Main structure of LSTM neurons.

the previous input. First, the previous hidden state h_{t-1} and the current input x_t is passed into the sigmoid function after which the newly obtained cell state is passed to the tanh function. Finally, the output of tanh is multiplied by the output of the sigmoid function to determine the information that the hidden state should carry. The hidden state h_t is then used as the output of the current cell, and the new cell state and new hidden state are passed to the next time step.

$$O_t = \sigma(w_o \cdot [h_{t-1}, x_t] + b_o), \quad (8)$$

$$h_t = O_t \odot \tanh(C_t), \quad (9)$$

where O_t denotes the output gate; w_o and b_o denote the weights and biases of the output gate, respectively; and h_t is the hidden state of the current layer.

To build the model, the original dataset was split into a training dataset (80%) and a testing dataset (20%). The performance metrics of the model were derived from both testing and external validation datasets. To prevent overfitting problems in the model training, a five-fold cross-validation of the training data was used to set the hyperparameters and model structure (Yan et al., 2019). Different model hyperparameters were set to obtain the best prediction results. The hyperparameters were set as listed in Table 1. The number of neurons and hidden layers determines the complexity of the model. Dropout was temporarily discarded from the network

according to a certain probability to prevent overfitting. The optimisation function was selected by mainly considering the decay of the learning rate from deciding the step size of the gradient descent to reduce the prediction error quantified by the loss function. The batch parameter was used to set the model to predict 1-, 6-, 12-, and 24-h data.

2.4. Explainable predictions

To better understand the features driving a specific prediction, Shapley additive explanations (SHAP) were applied to the proposed prediction model to obtain a visual interpretation of the features driving the prediction. SHAP is a model-independent representation of feature importance in which the influence of each feature on a particular prediction is represented by a game-theory-inspired Shapley value (Butnariu & Kroupa, 2008; Nowak & Radzik, 1994). Considering the current set of feature values, the Shapley value indicates the extent to which the variation between the actual and average predictions caused by any one feature under the influence of other features will contribute to the difference between the actual and average predictions. Hence, the sum of the Shapley values of all features and the average predicted value is equal to the actual predicted value. Importantly, this differs from the known direct feature effect in the (generalised) linear model. The Shapley values require that the results of each possible

Table 1 – Hyperparameter settings for the prediction model.

	Neurons	Hidden layer	Dropout coefficient	Optimisation function	Batch	Activation function
Parameter	1	1	0.0	RMSprop	1	Tanh
	4	2	0.1	Adagrad	6	Softsign
	8	3	0.2		12	
	16	5	0.4		24	
	32	10	0.6			
	64					
	128					

combination (or coalition) of elements be considered to determine the importance of each feature and resolve its potential dependencies.

The contribution of a given feature is determined, including the minutiae contribution and in all combinations of features (and association models) that do not contain, and then averaged. The Shapley values afford a unique approach for assigning the contribution or importance of each element that satisfies certain desired attributes. In particular, the sum of the Shapley values for each feature corresponds to the difference between the predicted value of the model and that of the virtual model without the feature. The Shapley value for each feature is determined as follows:

$$\phi_i(f, x) = \sum_{z' \subseteq x'} \frac{|z'|!(M - |z'| - 1)!}{M!} [f_x(z') - f_x(z' / i)], \quad (10)$$

where f denotes the prediction model, x is the input sample data, z' is a subset of the feature combinations, x' is a simplification of the input data, and M is the total number of features.

2.5. Model performance

The prediction accuracy metric measures the difference between the estimated value and its corresponding actual value. A common metric for evaluating the performance of deep learning-based imputation models is the root-mean-square error (RMSE), which reflects the degree of deviation between the true and predicted values. A smaller RMSE value indicates better model quality and more accurate predictions (Chicco et al., 2021; Malsa et al., 2021). To further validate the prediction performance of the model for different datasets, the coefficient of determination (R^2) was used to evaluate the model. The RMSE and R^2 are defined as follows:

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}}, \quad (11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (12)$$

where N represents the number of samples, y_i is the true value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the sample mean.

3. Experimental results

3.1. Median lethal concentration of total ammonia nitrogen

In the ammonia nitrogen toxicity test, by setting an NH_3 concentration interval of 0.25 mg L^{-1} , the maximum TAN concentration for no mortality and the minimum TAN concentration for complete mortality within 24 h were obtained as 23.12 mg L^{-1} and 46.34 mg L^{-1} , respectively. As summarised in Table 2, at a TAN concentration of 30.90 mg L^{-1} , the average fish mortality for 24 h was already as high as 23%; at a TAN concentration of 46.34 mg L^{-1} , all fish died within 24 h; and at a TAN concentration of 54.07 mg L^{-1} , all fish died within 12 h. Comparing ammonia nitrogen stress at adjacent gradient concentrations, fish mortality increased with increasing

concentration ($P < 0.05$). The number of fish mortalities at different ammonia nitrogen concentrations at 24, 48, 72, and 96 h were listed in Table 3. In general, fish mortality was positively correlated with the duration of ammonia nitrogen stress. No fish died when the TAN concentration was 22.91 mg L^{-1} for 24 h. However, when the stress lasted for 96 h, more than one-third of the fish died (mean mortality: 40%). The effect of stress duration on fish mortality was more significant ($P < 0.05$) when the TAN concentration was higher than 30.20 mg L^{-1} . According to the mean mortality rate of American black bass under different ammonia nitrogen concentration stress at each time interval, the TAN LC_{50} for each time interval was obtained using probit regression analysis. Table 4 shows that the TAN LC_{50} for 24 h was 33.82 (95% confidence interval (CI) 29.78 – 38.82) mg L^{-1} , while the TAN LC_{50} for 96 h was 24.76 (CI 22.10 – 26.57) mg L^{-1} , with a difference of nearly 10 mg L^{-1} . This indicates that the lethal effect of ammonia nitrogen is closely related to the exposure time and concentration of ammonia nitrogen in experimental animals. The higher the concentration of ammonia nitrogen in the water and the longer the exposure time of fish, the higher the mortality of fish.

3.2. Model performance and explanation

3.2.1. Model performance

The proposed model had a higher predictive performance over time in terms of mortality prediction within 168 h of exposure to low-concentration ammonia nitrogen stress (Fig. 4). In the testing dataset, the initial R^2 of the model was 0.72 (CI 0.63–0.80), which increased to 0.84 (CI 0.77–0.91) after 168 h. The corresponding RMSEs were 0.20 (CI 0.12–0.28) and 0.31 (CI 0.22–0.40), respectively. An external validation dataset (consisting of statistics from 50 sets of breeding pools) was used to verify the stability and validity of the model. In general, depending on the RMSE or R^2 , the predictive power of the model increases with time, with an RMSE of 0.33 (CI 0.24–0.42) and R^2 of 0.75 (CI 0.67–0.83) at 0 h. The RMSE and R^2 were 0.35 (CI 0.26–0.44) and 0.79 (CI 0.71–0.87) after 48 h, 0.31 (CI 0.22–0.40) and 0.79 (CI 0.71–0.87) after 96 h, 0.29 (CI 0.20–0.38) and 0.80 (CI 0.72–0.88) after 144 h, and 0.30 (CI 0.21–0.39) and 0.81 (CI 0.73–0.89) at 168 h, respectively (Table 5).

Table 2 – Effects of different ammonia nitrogen concentrations on the mortality of American black bass under stress for 24 h.

	TAN (mg L^{-1})	NH_3 (mg L^{-1})	Mortality rate		
			Mean $\pm$ S.D. (%)	Mean difference(%)	Sig.
1	15.45	0.50	0.00 $\pm$ 0.00	–	–
2	23.12	0.75	0.00 $\pm$ 0.00	0.00	–
3	30.90	1.00	23.33 $\pm$ 2.89	23.33	0.005
4	38.62	1.25	61.67 $\pm$ 7.64	38.34	0.023
5	46.34	1.50	100.00 $\pm$ 0.00	38.34	0.013
6	54.07	1.75	100.00 $\pm$ 0.00	0.00	–
7	61.79	2.00	100.00 $\pm$ 0.00	0.00	–

Note: S.D. is the standard deviation, Mean difference is the mean difference in mortality rate of adjacent gradient concentrations, Sig. Is the significance of the mortality rate of adjacent gradient concentrations.

Table 3 – Effects of different ammonia nitrogen concentrations on the mortality of American black bass under stress from 24 to 96 h.

TAN (mg L ⁻¹)		NH ₃ (mg L ⁻¹)	Mortality rate				
				24 h	48 h	72 h	96 h
1	22.91	0.74	Mean ± S.D. (%)	0.00 ± 0.00	6.67 ± 5.77	33.33 ± 5.77	40.00 ± 0.00
			Mean difference (%)	—	6.67	26.67	36.67
			Sig.	—	0.183	0.015	0.184
2	26.30	0.85	Mean ± S.D. (%)	20.00 ± 0.00	26.67 ± 5.77	50.00 ± 0.00	56.67 ± 20.82
			Mean difference (%)	—	6.67	23.33	6.67
			Sig.	—	0.183	0.019	0.634
3	30.20	0.98	Mean ± S.D. (%)	26.67 ± 5.77	43.33 ± 5.77	70.00 ± 10.00	80.00 ± 0.00
			Mean difference (%)	—	16.67	26.67	10.00
			Sig.	—	0.037	0.015	0.22
4	34.67	1.12	Mean ± S.D. (%)	60.00 ± 0.00	83.33 ± 5.77	100.00 ± 0.00	100.00 ± 0.00
			Mean difference (%)	—	23.33	16.67	0.00
			Sig.	—	0.019	0.038	—
5	39.81	1.29	Mean ± S.D. (%)	66.67 ± 5.77	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
			Mean difference (%)	—	33.33	0.00	0.00
			Sig.	—	0.009	—	—
6	45.71	1.48	Mean ± S.D. (%)	100.00	100.00	100.00	100.00
			Mean difference (%)	0.00	0.00	0.00	0.00
			Sig.	—	—	—	—

Note: S.D. is the standard deviation, Mean difference is the mean difference in mortality rate of adjacent time intervals, Sig. Is the significance of the mortality rate at adjacent time intervals.

Table 4 – Median lethal and safe concentrations of ammonia nitrogen for American black bass.

		LC ₅₀				SC
		24 h	48 h	72 h	96 h	
TAN (mg L ⁻¹)	Estimated value	33.82	30.27	26.72	24.76	2.48
	CI	29.78–38.82	27.89–32.70	24.41–28.74	22.10–26.57	2.21–2.66
NH ₃ (mg L ⁻¹)	Estimated value	1.09	0.98	0.86	0.80	0.08
	CI	0.96–1.26	0.90–1.06	0.79–0.93	0.72–0.86	0.07–0.09

Note: CI is the 95% confidence interval.

3.2.2. Predictive visualisation

We visualised the contribution of each of the 32 features in this prediction model, which included both continuous and

binary input features in Fig. 5. Continuous features varied from low to high, whereas binary features were either present or absent. Each dot represents the effect of a feature on the

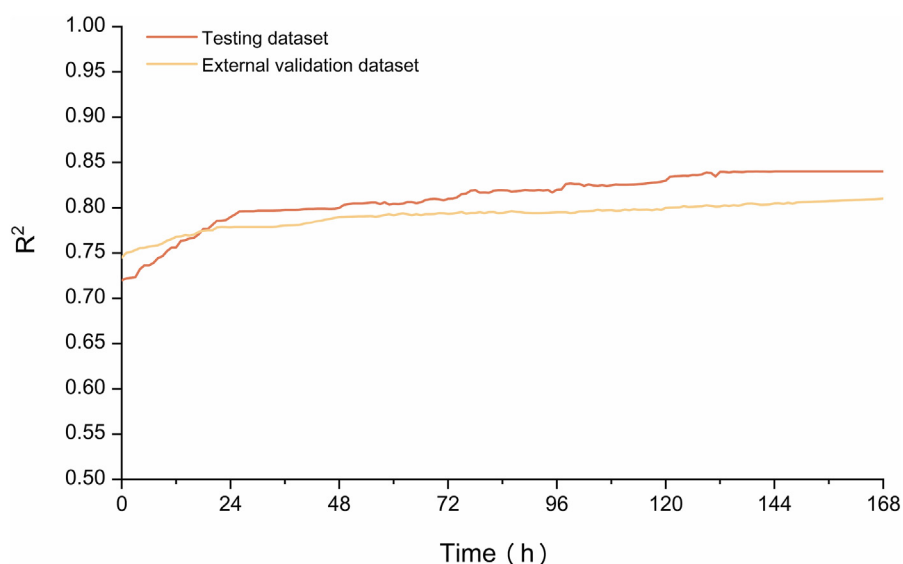
**Fig. 4 – Performance of the proposed prediction model.**

Table 5 – R^2 and RMSE of the prediction model.

	0 h	24 h	48 h	72 h	96 h	120 h	144 h	168 h
Testing dataset	R^2 (CI) 0.72 (0.63–0.80)	0.79 (0.71–0.87) 0.30 (0.21–0.39)	0.80 (0.72–0.88) 0.33 (0.24–0.42)	0.81 (0.73–0.89) 0.36 (0.27–0.45)	0.82 (0.74–0.90) 0.32 (0.23–0.41)	0.83 (0.76–0.90) 0.31 (0.22–0.40)	0.84 (0.77–0.91) 0.31 (0.22–0.40)	0.84 (0.77–0.91) 0.31 (0.22–0.40)
External validation dataset	R^2 (CI) 0.75 (0.67–0.83)	0.78 (0.70–0.86) 0.38 (0.28–0.48)	0.79 (0.71–0.87) 0.35 (0.26–0.44)	0.79 (0.71–0.87) 0.30 (0.21–0.39)	0.79 (0.71–0.87) 0.31 (0.22–0.40)	0.80 (0.72–0.88) 0.30 (0.21–0.39)	0.80 (0.72–0.88) 0.29 (0.20–0.38)	0.81 (0.73–0.89) 0.30 (0.21–0.39)

Note: CI is the 95% confidence interval.

predicted mortality of fish at a given point in time, with a darker shade of red representing a greater contribution. The dots on the left represent the feature values that push the prediction towards survival, and the dots on the right represent the feature values that push the prediction towards mortality.

The AVE_w of the fish had a significant effect on the predictions, with a larger AVE_w favouring the survival prediction and a smaller AVE_w favouring the non-survival prediction. This is consistent with the fact that body weight is an important factor that influences the degree of ammonia tolerance in fish (Section 3.1). In general, high values of most non-binary characteristics tend to influence mortality estimates in the direction of non-survival, whereas low values tend to influence it in the direction of survival. However, there are several exceptions to this rule. Larger values of the features of DO (maximum (DO_{max}), median (DO_{med}), and minimum (DO_{min})) typically drive mortality predictions towards survival. All binary features appear to affect the mortality predictions in a unidirectional manner (e.g., sterilisation or feeding pulls the prediction towards survival). NH₃ was the most toxic in the aqueous environment; however, under low-concentration ammonia nitrogen stress conditions, NH₃ ranked 24th out of the 32 characteristics within 168 h. This does not indicate that NH₃ is not harmful to fish. Rather, it indicates that a small concentration of NH₃ in the water does not pose a mortality threat to fish. However, NH₃ can significantly damage the immune system of aquatic animals and their respiratory metabolic systems, ultimately inhibiting growth and reproduction.

Dynamic mortality predictions can also be interpreted at any given time for a particular breeding pool. A representative case study is presented in Fig. 6. The red line indicates the mortality prediction of our model over time (i.e., higher values indicate higher mortality). The shaded area shows the three most important input features of non-survival (orange) and survival (blue) that drive the mortality prediction within 168 h. A high opacity reflects the high relative importance of the features; the numbers are used to identify the features, and labels are re-added when the importance of one feature exceeds that of another.

The AVE_w of the fish in this pond was 12.2 g, CD was 49.38 kg m⁻³, and the initial TAN concentration was 20.01 mg L⁻¹. The first feeding was performed after 6 h of ammonia stress at 2% of the total fish weight. After 18 h of stress, one-third of the total water volume in the culture pond was changed, and the TAN concentration in the water was reduced to 11.70 mg L⁻¹. After 60 h of stress, mortality began to occur in the culture pond; after 168 h of stress, the total number of mortalities in the culture pond was 24. In this case, the AVE_w value of the fish drove the survival prediction, whereas the DO_{max} and DO_{min} values were the most important features that mostly favoured the prediction towards survival during the stress period. In addition, the importance of hydrogen ion concentration maximum (pH_{max}) increased after approximately 60 h, driving the fish towards survival; however, after approximately 65 h, it drove them towards mortality. This was because the suitable pH environment for growth of most fish ranges from 6.5 to 8.5, and the rapid increase in pH causes them to be maladjusted in



Fig. 5 – Influence of each feature on the prediction of survival and death in fish.

the short term, which can hinder their growth or even lead to mortality. Initially, the water temperature median (T_med) drove the fish towards survival; however, at 61 h, it pushed the fish towards mortality, and after approximately 80 h, it pushed the fish towards survival. Water temperature maximum (T_max), hydrogen ion concentration difference (pH_diff), TAN, and NH₃ gradually pushed the fish towards

mortality. When considering the relative importance of all included features to the overall prediction, as shown in the mean ranking (Fig. 7), we noticed that the temperature and pH features were relatively high in importance throughout the time period. The importance of the DO features (relative to the other features) increased over time, whereas that of the medication operation decreased, which could be explained by

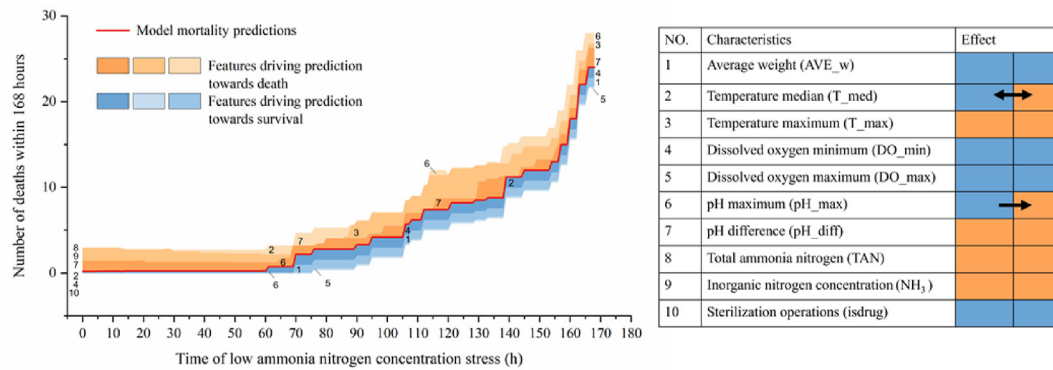


Fig. 6 – Effect of input features on dynamic mortality predictions for individual culture pools.

the fact that ammonia nitrogen is more damaging to the gills of fish. Therefore, maintaining DO is beneficial, as it improves fish respiration, whereas short periods of medication are not conducive to fish survival.

3.2.3. Comparison of different approaches

A comparison of the performance of the fish mortality prediction method in this study with that of previously reported methods is shown in Table 6. The prediction performance of the LSTM-based model was comparable to those of other multivariate linear models, principal component analysis (PCA), and random forest-based (RF) fish mortality prediction. Our method showed high prediction performance for the external validation dataset, with an R^2 value of 0.81. We selected multivariate data with a high correlation for further data fusion and fully considered the variation characteristics of each feature in the time series, which may be the reason for the excellent prediction performance of our method. Second,

the other three prediction methods are static. In other words, the prediction models rely only on the data input at the time of establishment, without considering the effect of new data on the prediction results during the development of the time series. In aquaculture prediction tasks, environmental data changes are dynamic and disordered, and these dynamically changing data often have a critical impact on the prediction results. LSTM can update the prediction by integrating new data and enhance the model through the temporal development of the features to improve the prediction performance.

From the perspective of model interpretability, this study proposes the use of game theory to determine the importance of each feature in each time interval by considering possible combined (or combined) outcomes of the elements. Unlike the direct feature effects known in linear models, this approach avoids the influence of covariance among the features on the prediction results (Dobriban & Wager, 2018; Li et al., 2021; Neumaier & Groeneveld, 1998) and enhances the

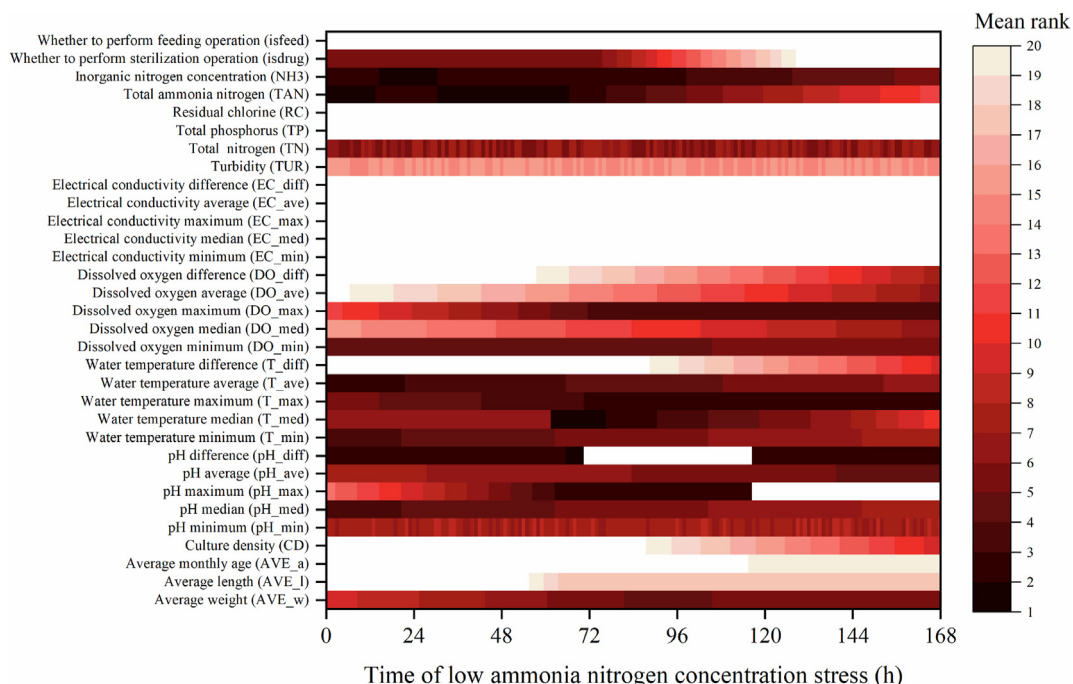


Fig. 7 – Mean ranking of the impact of input features on dynamic mortality for individual culture pools.

Table 6 – Comparison of different methods for predicting fish mortality.

Ref.	Prediction method	Data		Performance	Prediction type	Predicted step size	Model explanation
		Data type	Data selection				
Lopes et al. (2021)	PCA	8 water quality parameters	Not reported	F1-score 80%	Static	Daily	Abnormal probability
Inoue et al. (2020)	RF	5 marine environmental data and 4 fishery information variables	Not reported	Average accuracy 0.699	Static	Monthly	Loss value of variable
Jensen (2001)	Multiple linear regression	4 parameters (weight, length, natural growth rate of fish, and temperature)	P-Value	R ² 0.71	Static	Not reported	Correlation of variable
Our	LSTM + SHAP	32 data including static features and dynamic features	P-Value	R ² 0.81	Dynamic	Hourly	Effect of input features

interpretability of the model prediction results. Compared with other model interpretation mechanisms based on anomaly probabilities and model loss values, feature importance provides a more intuitive picture of the effect of changes in each parameter in the time series on fish mortality, and the prediction results are more comprehensible to decision makers.

4. Discussion

In an actual culture environment, fish may be exposed to low levels of ammonia stress for long periods. Most fish mortality is caused by excess ammonia in water. In this work, an ammonia nitrogen toxicity test was first conducted to determine the LC₅₀ and SC of ammonia nitrogen in American black bass. Sun (2021) studied the ammonia nitrogen LC₅₀ for American black bass weighing 20.2 ± 1.27 g. The fish species and water quality parameters for rearing used in their study were the same as those in our study; however, the median lethal concentration of total ammonia nitrogen within 24 h (24-h TAN LC₅₀) for the fish was 76.64 mg L^{-1} and the median lethal concentration of inorganic nitrogen (NH₃ LC₅₀) was 2.48 mg L^{-1} (Sun, 2021), both of which were higher than those used in this study (Table 4). The reason for this may be that, in addition to the effect of water temperature, the weight of the fish is one of the main factors influencing the ammonia emission rate (Paudel et al., 2015); the higher the weight of the fish, the higher the ammonia emission. The weight of the fish and ammonia emissions were correlated using a power function. Second, higher fish weights correspond to a stronger ability to convert NH₃ into less toxic substances such as glutamine and urea (Song et al., 2022). When excessive toxic glutamate is produced in fish owing to high ammonia nitrogen, the brain tissue activates a large amount of glutamine synthetase to convert glutamate and NH₄⁺ into non-toxic glutamine to protect the organism from dying. The species of the fish also affects its ammonia nitrogen tolerance. Among freshwater fish, juvenile crucian carp (*Carassius auratus*) had 96-h TAN LC₅₀ of 148.00 mg L^{-1} at pH 7.2, T 25 °C, and DO > 7.0 mg L⁻¹ (Hao et al., 2019). Juvenile grass carp (*Ctenopharyngodon idella*) had 96-h TAN LC₅₀ of 48.16 mg L^{-1} at pH 8.0, T 24 °C, and DO > 5.0 mg L⁻¹ (Cao et al., 2021). Juvenile Chinese perch (*Siniperca chuatsi*) was less tolerant to ammonia nitrogen, with 96-h TAN LC₅₀ of 8.61 mg L^{-1} at pH 7.5, T 21 °C, and DO > 8.0 mg L⁻¹ (Fang et al., 2017). Among marine fish, the 96-h TAN LC₅₀ of turbot juveniles (*Scophthalmus maximus*) was 61.95 mg L^{-1} at pH 8.0, T 23 °C, and DO > 5.0 mg L⁻¹ (le Ruyet et al., 1998). The 96-h TAN LC₅₀ of black seabream juveniles (*Acanthopagrus schlegelii*) was 30.00 mg L^{-1} at pH 8.1, T 25 °C, and DO 7.6 mg L⁻¹ (Park et al., 2015), whereas that of large yellow croaker juveniles (*Pseudosciaena crocea*) was 12.48 mg L^{-1} at pH 7.4, T 23 °C, and DO > 5.0 mg L⁻¹ (Cai et al., 2016). Overall, the 96-h TAN LC₅₀ of freshwater fish was higher than that of marine fish, which may be because the living space of marine fish is larger than that of freshwater fish. The smaller living space of freshwater fish makes them relatively more vulnerable to ammonia nitrogen pollution. In contrast, the concentration of ammonia nitrogen in marine water is relatively lower, resulting in lower tolerance of marine fish to ammonia nitrogen. Another factor is the salinity; the higher the salinity of

seawater, the smaller the proportion of NH_4^+ , the less the accumulation of ammonia nitrogen in the fish, and the easier it is to eliminate ammonia nitrogen from the body, which can reduce the toxic effect of ammonia nitrogen on fish. LC_{50} is an important parameter commonly used in biostatistics to measure the magnitude of toxicity of a toxicant (Neely, 1984; Tancredo et al., 2019). The lethal effects of toxins are closely related to the exposure time of the experimental animals. Further, the toxic effect of ammonia nitrogen concentration on fish in aquaculture water is easily affected by the T, pH, salinity, DO, and other environmental factors. Second, differences in fish species, size, age, and water quality can also affect the ammonia nitrogen tolerance of fish and, consequently, the concentration deemed semi-lethal to the fish. However, the ammonia nitrogen toxicity test is performed under conditions of strict control of water parameters; in actual breeding environments, parameters such as temperature and pH often change dynamically. Therefore, actual fish mortality in aquaculture is difficult to assess using toxicity tests.

Owing to the complexity of aquaculture environments, few studies have proposed fish mortality prediction models with dynamic and interpretable prediction results. This study presents a method for predicting fish mortality under low-concentration ammonia nitrogen stress based on a combination of time-recursive neural networks and game theory. The model performance results shown in Fig. 4 and Table 5 indicate that LSTM exhibits excellent prediction performance that improves with time. However, in the initial stage (0–24 h), the R^2 of the model was slightly better on the external validation dataset than on the testing dataset due to parameter pre-training. In general, the performance of the model on the external validation dataset was slightly worse than that on the test dataset. The environmental conditions of the breeding pools in the two datasets differed. For example, the number of ponds with CD greater than 50 kg m^{-3} in the testing dataset exceeded 98%, whereas that in the external validation dataset was only 52%. Hence, a decrease in the model prediction performance was expected. In comparison with static predictive models (Table 6), this approach allows the model to learn not only from the actual values of the variables at a particular time, but also from the temporal trends of the data obtained at each hour.

In addition, the visualisation of the contribution of 32 features in this prediction model shows the influence of each feature in predicting fish survival and mortality (Fig. 5). Furthermore, the model provides an hourly interpretation of the features that drive the predictions for individual culture pool. We observed that one feature could drive the predicted survival at one point in time and non-survival at another (Figs. 6 and 7). In comparison with explanatory mechanisms based on anomalous probabilities, functional loss values, and variable correlation indices (Table 6), this comprehensible model explanation mechanism allows farmers to combine temporal predictions with their existing expertise to facilitate decision making. To the best of our knowledge, this is the first time that a deep learning model has provided explanations in a longitudinal manner as farming progressed. This temporal ranking of features may help breeders decide and implement interventions at the time of breeding.

5. Conclusion

Fish in aquaculture environments may be exposed to low-concentration ammonia nitrogen stress for long periods of time owing to the uncertainty of environmental factors. Therefore, we developed a prediction model applicable to fish mortality under low-concentration ammonia nitrogen stress to predict the dynamics of fish mortality in stressful environments for 168 h. The model was trained on 32 binary and continuous environmental variables for 100 culture ponds and updated hourly. The performance of the model improved over time, achieving an R^2 value of 0.81 (CI 0.73–0.89) and RMSE of 0.30 (CI 0.21–0.39) on an external validation dataset. The model is interpretable and identifies important features that drive mortality prediction in both testing and external validation datasets. Importantly, in the analysis of mortality predictions for a given breeding pool, we observed that a feature could drive the predicted survival at one point in time and non-survival at another. None of these observations can be obtained from current static prognostic scores. However, because environmental tolerance varies among different species of fish, the application of the proposed method to other species of fish requires the collection of mortality data for that species and retraining of the model to achieve good performance.

In summary, this study identified the importance of a continuously updated decision support tool that can adapt to changing farming conditions and provide real-time guidance to farm personnel. This dynamic and interpretable predictive approach may be more useful in various decision-making activities than the method of static scoring currently implemented, for instance, in determining whether to initiate interventions and how to actively overcome environmental factors to drive fish survival.

CRediT authorship contribution statement

Yao Wu: Conceptualization, Methodology, Software, Data curation, Writing - original draft. **Xiaochan Wang:** Formal analysis, Supervision, Writing - review & editing, Funding acquisition. **Lin Wang:** Writing -review & editing. **Xiaolei Zhang:** Validation, Writing - review & editing. **Yinyan Shi:** Writing -review & editing. **Ye Jiang:** Writing -review & editing.

Data availability

The datasets created during and/or analysed during the current study are available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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